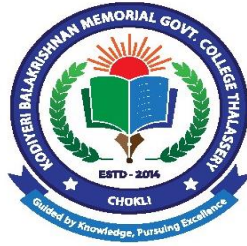


**KODIYERY BALAKRISHNAN MEMORIAL
GOVERNMENT COLLEGE THALASSERY**



Affiliated to KANNUR UNIVERSITY

**PG DEPARTMENT OF COMPUTER SCIENCE
2023-2024**

**FIRST SEMESTER M.Sc. COMPUTER SCIENCE
WITH SPECIALIZATION IN ARTIFICIAL
INTELLIGENCE**

MSCSC01CO6 LAB-I (C/C++/DS)

NAME:

REG NO :

**PG DEPARTMENT OF COMPUTER SCIENCE
2023-2024**

KODIYERI BALAKRISHNAN MEMORIAL GOVERNMENT COLLEGE THALASSERY

Affiliated to KANNUR UNIVERSITY



PG DEPARTMENT OF COMPUTER SCIENCE

2023-2024

CERTIFICATE

This is to certify that this bonafide record of the original work done by
Ms/Mr. _____ Reg no _____ of First semester
M.Sc. Computer Science With Specialization in Artificial Intelligence in
the Introduction to Programming and C,C++, Data Structure lab during the
year 2023- 2024.

Head of the Department :

Lecture in Charge:

Submitted for the university Examination held on.....

Examiner 1:

Examiner 2:

INDEX

C PROGRAM

SL.NO	PROGRAM NAME	PAGE.NO	REMARK
1	Simple if , if-else , Nested if , elseif ladder , switch		
2	while , do-while and for , nested loops		
3	One dimensional and Two dimensional arrays		
4	Functions , Recursion		
5	Structures and unions		
6	Pointers		
7	Files		
8	Command Line Arguments		
9	Preprocessors		

C++ PROGRAMS

SL.NO	PROGRAMME NAME	PAGE.NO	REMARK
1	inline Functions		
2	new and delete		
3	Function Overloading		
4	Friend Function and class		
5	Classes and objects		
6	Constructors-Parameterised Constructor , Multiple Constructor in a Class , Copy Constructor and Destructors		
7	Operator overloading-Binary and Unary		
8	Inheritance-Single , Multiple, Multilevel, Hierarchical and hybrid		
9	Virtual base classes , Virtual Functions		
10	C++ Files		
11	Class Templates and Function Templates		

DATA STRUCTURES

SI.NO	PROGRAM NAME	PAGE NO.	REMARK
1	Linked List Operations		
2	Polynomial Addition Using Arrays		
3	Sparse Matrix Addition Using Arrays		
4	Stack Using Linked List		
5	Queue Operations Using Arrays		
6	Evaluation Of Arithmetic Expressions		
7	BST Operations		
8	Heap Sort		
9	Hashing		
10	Binary Search		
11	Quick Sort		
12	N-queen Problem		

C PROGRAMS

1.Simple if

```
#include<stdio.h>
int main()
{
    int a,b;
    printf("Enter two numbers :");
    scanf("%d %d",&a,&b);
    if(a<b)
    {
        printf("%d is smaller than %d",a,b);
    }
    return 0;
}
```

OUTPUT

Compile Result

```
Enter two numbers :5 6
5 is smaller than 6
[Process completed - press Enter]
```

If else

```
#include<stdio.h>

int main()
{
    int n=0;
    printf("Enter a number:");
    scanf("%d",&n);
    if(n%2==0){
        printf("%d is even number",n);
    }
    else
    {
        printf("%d is odd number",n);
    }
    return 0;
}
```

OUTPUT

Compile Result

```
Enter a number:6  
6 is even number  
[Process completed - press Enter]
```

nested if

```
#include <stdio.h>

int main()
{
    int n1, n2, n3;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &n1, &n2, &n3);
    if(n1 > n2)
    {
        if(n1 > n3)
        {
            printf("%d is the maximum",n1);
        }
        else
        {
            printf("%d is the maximum",n3);
        }
    }
    else
    {
        if(n2 > n3)
        {
            printf("%d is the maximum",n2);
        }
        else
```

```
{  
printf("%d is the maximum",n3);  
}  
}  
return 0;  
}
```

OUTPUT

Compile Result

```
Enter three numbers: 2 10 5  
10 is the maximum  
[Process completed - press Enter]
```

else if ladder

```
#include<stdio.h>

int main()
{
    int m;
    printf("enter the mark :");
    scanf("%d",&m);
    if(m>=90)
    {
        printf("\nA+ Grade");
    }
    else if(m>=80)
    {
        printf("\nA Grade");
    }
    else if(m>=70)
    {
        printf("\n B Grade");
    }
    else if(m>=60)
    {
        printf("\nC Grade");
    }
    else if(m>=50)
```

```
{  
    printf("\nD Grade");  
}  
else if(m>=40)  
{  
    printf("\nE Grade");  
}  
else if(m<40)  
{  
    printf("\nfailed");  
}  
else  
{  
    printf("\nwrong entry");  
}  
}
```

OUTPUT

Compile Result

enter the mark :80

A Grade

[Process completed - press Enter]

switch

```
#include<stdio.h>

void main()
{
    char c;
    printf("enter the character :");
    scanf("%c",&c);
    if((c>='A')&&(c<='Z')||(c>='a')&&(c<='z'))
    {
        printf("\nit is a character");
        switch(c)
        {
            case 'A':
            case 'E':
            case 'I':
            case 'O':
            case 'U':
            case 'a':
            case 'e':
            case 'i':
            case 'o':
            case 'u':printf("\ncharacter is vowel ");
                    break;
            default:printf("\ncharacter is consonant");
                    break;
        }
    }
}
```

```
}  
}  
}
```

OUTPUT

Compile Result

```
enter the character :A
```

```
it is a character
```

```
character is vowel
```

```
[Process completed (code 20) - press Enter]
```

2.while

```
#include<stdio.h>
int main()
{
    int n,b=0,r,i=1;
    printf("Enter a number :");
    scanf("%d",&n);
    while(n!=0)
    {
        r=n%2;
        b=b+r*i;
        n=n/2;
        i=i*10;
    }
    printf("\nbinary number=%d",b);
}
```

OUTPUT

Compile Result

```
Enter a number :5
```

```
binary number=101
```

```
[Process completed - press Enter]
```

for loop

```
#include <stdio.h>

int main()
{
    int n, count, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    for(count=1;count<=n;count++)
    {
        sum += count;
    }
    printf("Sum=%d", sum);
    return 0;
}
```

OUTPUT

Compile Result

```
Enter a positive integer: 5
Sum=15
[Process completed - press Enter]
```

nested loop

```
#include <stdio.h>

int main()
{
    int end ;
    printf("Pattern Printing");
    printf("\n-----");
    printf("\nEnter a number:");
    scanf("%d",&end);

    int i = 1;
    while (i <= end)
    {
        printf("\n");
        int j = 1;
        while (j <= i)
        {
            printf("%d ", j);
            j = j + 1;
        }
        i = i + 1;
    }
    return 0;
}
```

OUTPUT

Compile Result

Pattern Printing

Enter a number:5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

[Process completed - press Enter]

do while

```
#include<stdio.h>
int main()
{
int i=1,n;
printf("Enter the limit :");
scanf("%d",&n);
do
{
printf("%d \n",i);
i++;
}while(i<=n);
return 0;
}
```

OUTPUT

Compile Result

```
Enter the limit :10
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
6
```

```
7
```

```
8
```

```
9
```

```
10
```

```
[Process completed - press Enter]
```

3.One dimensional array

```
#include<stdio.h>

int main()
{
    int a[50],limit,i;
    printf("Enter the limit :");
    scanf("%d",&limit);
    for(i=0; i<limit; i++)
    {
        printf("\nEnter the element :");
        scanf("%d",&a[i]);
    }
    for(i=0; i<limit; i++)
    {
        printf("%d\t",a[i]);
    }
}
```

OUTPUT

Compile Result

```
Enter the limit :4
Enter the element :1
Enter the element :4
Enter the element :3
Enter the element :7
1      4      3      7
[Process completed - press Enter]
```

Two dimensional array

```
#include<stdio.h>

int main()
{
    int a[5][5],b[5][5],c[5][5],r1,c1,r2,c2,i,j,k;
    printf("Matrix Multiplication\n");
    printf("\nEnter the row and column size of first matrix:\n");
    scanf("%d %d",&r1,&c1);
    printf("\nEnter the row and column size of second matrix :\n");
    scanf("%d %d",&r2,&c2);
    if(c1==r2)
    {
        printf("\nEnter the elements of first matrix\n");
        for(i=0; i<r1; i++)
        {
            for(j=0; j<c1; j++)
            {
                scanf("%d",&a[i][j]);
            }
        }
        printf("\nEnter the elements of second matrix\n");
        for(i=0; i<r2; i++)
        {
            for(j=0; j<c2; j++)
```

```

        {
            scanf("%d",&b[i][j]);
        }
    }
    for(i=0; i<r1; i++)
    {
        for(j=0; j<c2; j++)
        {
            c[i][j]=0;
            for(k=0; k<c1; k++)
            {
                c[i][j]=c[i][j]+a[i][k]*b[k][j];
            }
        }
    }
    printf("\nproduct matrix is \n");
    for(i=0; i<r1; i++)
    {
        printf("\n");
        for(j=0; j<c2; j++)
        {
            printf("%d ",c[i][j]);
        }
    }
}
else

```

```
{  
    printf("\nmatrix multiplication is not possible \n");  
}  
}
```

OUTPUT

Compile Result

Matrix Multiplication

Enter the row and column size of first matrix:

2 2

Enter the row and column size of second matrix :

2 2

Enter the elements of first matrix

1 2 3 4

Enter the elements of second matrix

1 2 3 4

product matrix is

7 10

15 22

[Process completed - press Enter]

4.Function

```
#include<stdio.h>
float sum(float,float);
int main()
{
    float a,b,s1,s;
    s1=sum(5,8);
    printf("%.2f",s1);
}
float sum(float a,float b)
{
    return(a+b);
}
```

OUTPUT

Compile Result

```
13.00
[Process completed - press Enter]
```

Recursion

```
#include<stdio.h>
int binomial(int,int);
int main()
{
    int n,r;
    printf("\nEnter the value of n and r:");
    scanf("%d %d",&n,&r);
    printf("\nbinomial coefficient =%d",binomial(n,r));
}
int binomial(int n,int r)
{
    if(r==0||r==n)
        return 1;
    else
        return(binomial(n-1,r-1)+binomial(n-1,r));
}
```

OUTPUT

Compile Result

```
Enter the value of n and r:5 2
```

```
binomial coefficient =10
```

```
[Process completed - press Enter]
```

5.Structures

```
#include<stdio.h>

struct class
{
int number;
char name[30];
float mark;
};

void main()
{
struct class s1={222,"ram",65};
struct class s2={333,"reddy",57};
struct class s3;

int x;

s3=s2;

x=((s3.number==s2.number)&&(s3.mark==s2.mark))?1:0;

if(x!=0)
{
printf("\n Structures s2 and s3 are equal");
printf("\n %d \t%s \t%f\n",s3.number,s3.name,s3.mark);
}
else
{
printf("\n Structures s2 and s3 are not equal");
```

```
}  
}
```

OUTPUT

Compile Result

```
Structures s2 and s3 are equal  
333    reddy    57.000000
```

```
[Process completed (code 24) - press Enter]
```

Union

```
#include<stdio.h>

union Myunion
{
int integer;
float floating_point;
};

int main()
{
union Myunion data;
data.integer=42;
printf("value as an integer:%d\n",data.integer);
data.floating_point=3.14;
printf("values as a floating point %f\n",data.floating_point);
}
```

OUTPUT

Compile Result

```
value as an integer:42  
values as a floating point 3.140000  
  
[Process completed - press Enter]
```

6.Pointers

```
#include<stdio.h>

int main()
{
int num[]={1,3,5,7,9};
int *ptr,i;
printf("\nThe array values are:\n");
for(i=0;i<5;i++)
{
printf("%d\t",num[i]);
}
ptr=num;
printf("\n value of ptr:%d",*ptr);
ptr++;
printf("\n value of ptr++:%d",*ptr);
ptr--;
printf("\n value of ptr--:%d",*ptr);
ptr=ptr+2;
printf("\n value of ptr+2:%d",*ptr);
ptr--;
printf("\n value of ptr--:%d",*ptr);
ptr=ptr+3;
printf("\n value of ptr+3:%d",*ptr);
printf("\n");
```

```
}
```

OUTPUT

Compile Result

The array values are:

1 3 5 7 9

value of ptr:1

value of ptr++:3

value of ptr--:1

value of ptr+2:5

value of ptr--:3

value of ptr+3:9

[Process completed - press Enter]

7.Files

```
#include<stdio.h>

int main()
{
    FILE *file;
    char data[100];
    file=fopen("abc.txt","w");
    if(file==NULL)
    {
        printf("Error opening file for writing.\n");
        return 1;
    }
    printf("Enter data to write to the file:");
    fgets(data,sizeof(data),stdin);
    fprintf(file,"%s",data);
    fclose(file);
    file=fopen("abc.txt","r");
    if(file==NULL)
    {
        printf("Error opening file for reading\n");
        return 1;
    }
    printf("Data read from the file :");
    while(fgets(data,sizeof(data),file)!=NULL)
```

```
{  
    printf("%s",data);  
}  
fclose(file);  
return 0;  
}
```

OUTPUT

Compile Result

```
Enter data to write to the file:Hello  
Data read from the file :Hello  
  
[Process completed - press Enter]
```

8.Command Line Arguments

```
#include <stdio.h>

int main(int argc, char* argv[])
{
    printf("Program name is: %s", argv[0]);
    if (argc == 1)
        printf("\nNo Extra Command Line Argument Passed ""Other Than Program Name");
    if (argc >= 2)
    {
        printf("\nNumber Of Arguments Passed: %d", argc);
        printf("\n----Following Are The Command Line " "Arguments Passed----");
        for (int i = 0; i < argc; i++)
            printf("\nargv[%d]: %s", i, argv[i]);
    }
    return 0;
}
```

OUTPUT

Compile Result

```
Program name is: /data/user/0/com.kvassyu.coding2.c/files/default  
No Extra Command Line Argument Passed Other Than Program Name  
[Process completed - press Enter]
```

9.Preprocessors

```
#include<stdio.h>

#define ADDITION(x,y) ((x)+(y))
#define SUBTRACTION(x,y) ((x)-(y))
#define MULTIPLICATION(X,Y) ((X)*(Y))
#define DIVISION(x,y) ((y)!=0?(float)(x)/(float)(y):printf("cannot dividsible\n"))

int main()
{
    int n1,n2;
    printf("Enter two numbers:");
    scanf("%d %d",&n1,&n2);
    printf("sum: %d\n",ADDITION(n1,n2));
    printf("difference : %d\n",SUBTRACTION(n1,n2));
    printf("product : %d\n",MULTIPLICATION(n1,n2));
    printf("division : %f\n",DIVISION(n1,n2));
}
```

OUTPUT

Compile Result

```
Enter two numbers:4 2  
sum: 6  
difference : 2  
product : 8  
division : 2.000000
```

```
[Process completed - press Enter]
```

CPP PROGRAMS

1.Inline Functions

```
#include<iostream>
using namespace std;
class operation
{
    int a,b,add,sub;
public:
    void get();
    void sum();
    void difference();
};
inline void operation::get()
{
    cout<<"Enter first value :";
    cin>>a;
    cout<<"Enter second value :";
    cin>>b;
}
inline void operation::sum()
{
    add=a+b;
    cout<<"Addition of two numbers:"<<a+b<<"\n";
}
inline void operation::difference()
```

```
{
    sub=a-b;
    cout<<"Difference of two numbers :"<<a-b<<"\n";
}
int main()
{
    cout<<"Program using inline function\n";
    operation s;
    s.get();
    s.sum();
    s.difference();
}
```

OUTPUT

Compile Result

```
Program using inline function
Enter first value :6
Enter second value :2
Addition of two numbers:8
Difference of two numbers :4
```

```
[Process completed - press Enter]
```

2.new and delete

```
#include <iostream>
using namespace std;
int main()
{

    int num;
    cout << "Enter total number of students: ";
    cin >> num;
    float* ptr;
    ptr = new float[num];
    cout << "Enter GPA of students." << endl;
    for (int i = 0; i < num; ++i)
    {
        cout << "Student" << i + 1 << ": ";
        cin >> *(ptr + i);
    }
    cout << "\nDisplaying GPA of students." << endl;
    for (int i = 0; i < num; ++i)
    {
        cout << "Student" << i + 1 << ": " << *(ptr + i) << endl;
    }
    delete[] ptr;
    return 0;
}
```

OUTPUT

Compile Result

```
Enter total number of students: 4
```

```
Enter GPA of students.
```

```
Student1: 8.9
```

```
Student2: 8.7
```

```
Student3: 7.8
```

```
Student4: 8.5
```

```
Displaying GPA of students.
```

```
Student1: 8.9
```

```
Student2: 8.7
```

```
Student3: 7.8
```

```
Student4: 8.5
```

```
[Process completed - press Enter]
```

3.Function Overloading

```
#include <iostream>
using namespace std;
void add(int a, int b)
{
    cout << "\nsum = " << (a + b);
}
void add(int a, int b, int c)
{
    cout << "\nsum = " << (a + b + c);
}
int main()
{
    add(10, 2);
    add(5, 6, 4);
    return 0;
}
```

OUTPUT

Compile Result

```
sum = 12  
sum = 15  
[Process completed - press Enter]
```

4.Friend Function and Class

```
#include <iostream>
using namespace std;
class Example2;
class Example1
{
    private:
        int n1;
    public:
        Example1()
        {
            n1 = 20;
        }
        friend class Example2;
        friend void square(Example1 num);
};
void square(Example1 num)
{
    cout << "friend function\n square=" << num.n1*num.n1<<"\n";
}
class Example2
{
    private:
        int n2;
```

```
public:
void display(Example1& i)
{
    cout<<"friend class\nvalue="<<i.n1<<"\n";
}
};

int main()
{
    Example1 ob;
    Example2 n2;
    n2.display(ob);
    square(ob);
    return 0;
}
```

OUTPUT

Compile Result

```
friend class  
value=20  
friend function  
square=400
```

```
[Process completed - press Enter]
```

5.Classes and Object

```
#include <iostream>
using namespace std;
class Employee
{
public:
    int id;
    string name;
    float salary;
    void insert(int i, string n, float s)
    {
        id = i;
        name = n;
        salary = s;
    }
    void display()
    {
        cout<<id<<" "<<name<<" "<<salary<<"\n";
    }
};
int main(void)
{
    Employee e1;
```

```
Employee e2;  
e1.insert(101, "Abc",99000);  
e2.insert(102, "Ghi", 29000);  
e1.display();  
e2.display();  
return 0;  
}
```

OUTPUT

Compile Result

```
101   Abc  99000  
102   Ghi  29000
```

```
[Process completed - press Enter]
```

6. Constructors

```
#include<iostream>
using namespace std;
class sum
{
    int num1,num2,result;
public:
    sum()
    {
        cout<<"enter 2 numbers:";
        cin>>num1>>num2;
        result=num1+num2;
    }
    void display()
    {
        cout<<"result="<<result<<endl;
    }
};
int main()
{
    sum s;
    s.display();
}
```

OUTPUT

Compile Result

```
enter 2 numbers:5 8  
result=13
```

```
[Process completed - press Enter]
```

Parameterized Constructors

```
#include<iostream>
using namespace std;
class student
{
    float mark;
    int age;
public:
    student(float a,float b)
    {
        mark=a;
        age=b;
    }
    void display()
    {
        cout<<"mark="<<mark<<endl;
        cout<<"age="<<age<<endl;
    }
};
int main()
{
    student s(40.5,20);
    s.display();
}
```

OUTPUT

Compile Result

```
mark=40.5  
age=20
```

```
[Process completed - press Enter]
```

Multiple Constructors in a Class

```
#include <iostream>
using namespace std;
class construct
{
public:
    float area;
    construct()
    {
        area = 0;
    }
    construct(int a, int b)
    {
        area = a * b;
    }
    void display ()
    {
        cout<< area<< endl;
    }
};

int main()
{
    construct c1;
    construct c2( 10, 20);
```

```
c1.display();  
c2.display();  
}
```

OUTPUT

Compile Result

```
0  
200
```

```
[Process completed - press Enter]
```

Copy Constructor

```
#include<iostream>
using namespace std;
class Sample
{
    int id;
public:
    void init(int x)
    {
        id=x;
    }
    Sample()
    {
    }
    Sample(Sample &t)
    {
        id=t.id;
    }
    void display()
    {
        cout<<"\nID="<<id;
    }
};
int main()
```

```
{  
    Sample obj1;  
    obj1.init(10);  
    obj1.display();  
    Sample obj2(obj1);  
    obj2.display();  
}
```

OUTPUT

Compile Result

```
ID=10  
ID=10  
[Process completed - press Enter]
```

Destructor

```
#include <iostream>
using namespace std;
class Test
{
public:
    Test()
    {
        cout << "\n Constructor executed";
    }
    ~Test()
    {
        cout << "\n Destructor executed";
    }
};
int main()
{
    Test t,t2;
}
```

OUTPUT

Compile Result

```
Constructor executed  
Constructor executed  
Destructor executed  
Destructor executed  
[Process completed - press Enter]
```

7.Operator Overloading

Binary Operator Overloading

```
#include<iostream>
using namespace std;
class complex
{
    float x;
    float y;
public:
    complex()
    {
    }
    complex(float real,float img)
    {
        x=real;
        y=img;
    }
    complex operator+(complex);
    void display();
};
complex complex::operator+(complex c)
{
    complex temp;
```

```

    temp.x=x+c.x;
    temp.y=y+c.y;
    return(temp);
}
void complex::display()
{
    cout<<x<<" +i"<<y<<"\n";
}
int main()
{
    complex c1,c2,c3;
    c1=complex(2.5,3.5);
    c2=complex(1.5,2.5);
    c3=c1+c2;
    cout<<"c1=";
    c1.display();
    cout<<"c2=";
    c2.display();
    cout<<"c3=";
    c3.display();
    return 0;
}

```

OUTPUT

Compile Result

```
c1=2.5+i3.5  
c2=1.5+i2.5  
c3=4+i6
```

```
[Process completed - press Enter]
```

Unary Operator Overloading

```
#include<iostream>
using namespace std;
class box
{
private:
    int x;
public:
    box()
    {
        x=0;
    }
    box(int a)
    {
        x=a;
    }
    void print()
    {
        cout<<"X : "<<x<<"\n";
    }
    void operator ++()
    {
        ++x;
    }
}
```

```
}  
  
void operator ++(int)  
{  
    x++;  
}  
  
};  
  
int main()  
{  
    box b;  
    b.print();  
    ++b;  
    b.print();  
    b++;  
    b.print();  
}
```

OUTPUT

Compile Result

```
X : 0  
X : 1  
X : 2
```

```
[Process completed - press Enter]
```

8.Inheritance

Single Inheritance

```
#include<iostream>
using namespace std;
class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle\n";
    }
};
class Car : public Vehicle
{
};
int main()
{
    Car obj;
}
```

OUTPUT

Compile Result

```
This is a Vehicle
```

```
[Process completed - press Enter]
```

Multiple Inheritance

```
#include <iostream>
using namespace std;
class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle\n";
    }
};
class FourWheeler
{
public:
    FourWheeler()
    {
        cout << "This is a 4 wheeler Vehicle\n";
    }
};
class Car : public Vehicle, public FourWheeler
{
};
int main()
{
    Car obj;
```

```
}
```

OUTPUT

Compile Result

```
This is a Vehicle  
This is a 4 wheeler Vehicle  
[Process completed - press Enter]
```

Multi-level Inheritance

```
#include <iostream>
using namespace std;
class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle\n";
    }
};
class fourWheeler : public Vehicle
{
public:
    fourWheeler()
    {
        cout << "Objects with 4 wheels are vehicles\n";
    }
};
class Car : public fourWheeler
{
public:
    Car()
    {
        cout << "Car has 4 Wheels\n";
    }
};
```

```
    }  
};  
int main()  
{  
    Car obj;  
}
```

OUTPUT

Compile Result

```
This is a Vehicle  
Objects with 4 wheels are vehicles  
Car has 4 Wheels  
  
[Process completed - press Enter]
```

Hierarchical Inheritance

```
#include <iostream>
using namespace std;
class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle\n";
    }
};
class Car : public Vehicle
{
};
class Bus : public Vehicle
{
};
int main()
{
    Car obj1;
    Bus obj2;
}
```

OUTPUT

Compile Result

```
This is a Vehicle  
This is a Vehicle
```

```
[Process completed - press Enter]
```

Hybrid Inheritance

```
#include <iostream>
using namespace std;
class Vehicle
{
public:
    Vehicle()
    {
        cout << "This is a Vehicle\n";
    }
};
class Fare
{
public:
    Fare()
    {
        cout << "Fare of Vehicle\n";
    }
};
class Car : public Vehicle
{
};
class Bus : public Vehicle, public Fare
{
};
int main()
{
    Bus obj2;
}
```

OUTPUT

Compile Result

```
This is a Vehicle  
Fare of Vehicle
```

```
[Process completed - press Enter]
```

9.Virtual Base Class

```
#include <iostream>
using namespace std;
class A
{
public:
    int a;
    A()
    {
        a = 10;
    }
};
class B : public virtual A
{
};
class C : public virtual A
{
};
class D : public B, public C
{
};
int main()
{
    D object;
    cout << "a = " << object.a ;
}
```

OUTPUT

Compile Result

```
a = 10
```

```
[Process completed - press Enter]
```

Virtual Functions

```
#include<iostream>

using namespace std;

class base
{
public:
void dis()
{
cout<<"\n display base";
}

virtual void show()
{
cout<<"\n show base";
}

};

class derived:public base
{
void dis()
{
cout<<"\n display derived";
}

void show()
{
cout<<"\n show derived";
}

};
```

```
int main()
{
    base b;
    derived d;
    base *bptr;
    cout<<"\n BPTR POINTS TO BASE";
    bptr=&b;
    bptr->dis();
    bptr->show();
    cout<<"\n BPTR POINTS TO DERIVED";
    bptr=&d;
    bptr->dis();
    bptr->show();
    cout<<"\n\n";
    return 0;
}
```

OUTPUT

Compile Result

```
BPTR POINTS TO BASE  
display base  
show base  
BPTR POINTS TO DERIVED  
display base  
show derived
```

```
[Process completed - press Enter]
```

10. C++ Files

```
#include<iostream>
#include<fstream>
using namespace std;
int main()
{
    ofstream outf("ITEM");
    cout<<"Enter item name:";
    char name[30];
    cin>>name;
    outf<<name<<"\n";
    cout<<"Enter item cost:";
    float cost;
    cin>>cost;
    outf<<cost<<"\n";
    outf.close();
    ifstream inf("ITEM");
    inf>>name;
    inf>>cost;
    cout<<"\n";
    cout<<"item name:"<<name<<"\n";
    cout<<"item cost:"<<cost<<"\n";
    inf.close();
}
```

OUTPUT

Compile Result

```
Enter item name:Abc  
Enter item cost:50
```

```
item name:Abc  
item cost:50
```

```
[Process completed - press Enter]
```

11.Class Templates

```
#include <iostream>
using namespace std;
template <typename T>
class Array
{
private:
    T* ptr;
    int size;
public:
    Array(T arr[], int s);
    void print();
};
template <typename T> Array<T>::Array(T arr[], int s)
{
    ptr = new T[s];
    size = s;
    for (int i = 0; i < size; i++)
        ptr[i] = arr[i];
}
template <typename T> void Array<T>::print()
{
    for (int i = 0; i < size; i++)
        cout << " " << *(ptr + i);
```

```
        cout << "\n";  
    }  
int main()  
{  
    int arr[5] = { 1, 2, 3, 4, 5 };  
    Array<int> a(arr, 5);  
    a.print();  
}
```

OUTPUT

Compile Result

```
1 2 3 4 5
```

```
[Process completed - press Enter]
```

Function Templates

```
#include <iostream>
using namespace std;

template <class T> void bubbleSort(T a[], int n)
{
    for (int i = 0; i < n - 1; i++)
        for (int j = n - 1; i < j; j--)
            if (a[j] < a[j - 1])
                swap(a[j], a[j - 1]);
}

int main()
{
    int a[5] = { 10, 50, 30, 40, 20 };
    int n = sizeof(a) / sizeof(a[0]);
    bubbleSort<int>(a, n);
    cout << " Sorted array : ";
    for (int i = 0; i < n; i++)
        cout << a[i] << " ";
    cout << "\n";
}
```

OUTPUT

Compile Result

```
Sorted array : 10 20 30 40 50
```

```
[Process completed - press Enter]
```

DATA STRUCTURE

1.Linked List Operations

```
#include<iostream>
using namespace std;
class node
{
public:
int data;
struct node *next;
};
class list
{
private:
node *head,*fresh,*temp;
public:
list()
{
head=NULL;
}
void creation()
{
fresh= new node;
cout<<"Enter data to be
inserted:";
cin>>fresh->data;
fresh->next=NULL;
if(head==NULL)
{
head=temp=fresh;
```

```

}
else
{
temp->next=fresh;
temp=fresh;
}
}
void display()
{
cout<<"Elements of linked list are: ";
temp=head;
while(temp!=NULL)
{
cout<<temp->data<<" ";
temp=temp->next;
}
}
void insertatbeg()
{
fresh=new node;
cout<<"Enter value to be inserted:";
cin>>fresh->data;
fresh->next=head;
head=fresh;
}
void insertatend()
{
fresh=new node;
cout<<"Enter value to be inserted:";

```

```

cin>>fresh->data;
fresh->next=NULL;
temp=head;
while(temp->next!=NULL)
{
temp=temp->next;
}
temp->next=fresh;
}
void insertatpos()
{
int pos;
fresh=new node;
cout<<"Enter value to be inserted:";

cin>>fresh->data;
cout<<"Enter position to be inserted:";
cin>>pos;
int i=1;
temp=head;
while(i<pos-1)
{
temp=temp->next;
i++;
}
fresh->next=temp->next;
temp->next=fresh;
}
void delfrombeg()
{

```

```

if(head==NULL)
{
cout<<"List empty, nothing to delete";
}
else
{
temp=head;
cout<<head->data<<"is deleted";
head=head->next;
delete temp;
}}
void delfromend()
{
node *prevnode;
if(head==NULL)
{
cout<<"List empty, nothing to
delete";
}
else
{
temp=head;
while(temp->next!=NULL)
{
prevnode=temp;
temp=temp->next;
}
if(head==temp)
{

```

```

cout<<head->data<<"is deleted";
head=NULL;
delete temp;
}
else
{
cout<<temp->data<<"is deleted";
prevnode->next=NULL;
delete temp;
}
}
}
void delatpos()
{
node *nextnode;
int pos,i=1;
temp=head;
cout<<"Enter position to be deleted";
cin>>pos;
if(pos==1)
{
delfrombeg();
}
else
{
while(i<pos-1)
{
temp=temp->next;
i++;
}
}
}

```

```

nextnode=temp->next;
cout<<nextnode->data<<"is deleted";temp->next=nextnode->next;9
delete nextnode;
}
}
int main()
{
int ch1,ch2;
list l;
do
{
cout<<"\nLinked ListImplementation\n";
cout<<"Enter your choice:\n";
cout<<"1. Creation\n";
cout<<"2. Display\n";
cout<<"3. Insert at Begining\n";
cout<<"4. Insert at End\n";
cout<<"5. Insert at Position\n";
cout<<"6. Delete from Begining\n";
cout<<"7. Delete from End\n";
cout<<"8. Delete from a Position\n";
cin>>ch1;
switch(ch1)
{
case 1: l.creation();
        break;
case 2: l.display();
        break;
case 3: l.insertatbeg();

```

```
        break;
case 4: l.insertatend();
        break;
case 5: l.insertatpos();
        break;
case 6: l.delfrombeg();
        break;
7: l.delfromend();
        break;
case 8: l.delatpos();
        break;
default: cout<<"Please enter the correct choice";
}
cout<<"Do you want to continue:(Enter 1 for yes 0 for No)";
cin>>ch2;
} while(ch);
}
```

OUTPUT

Compile Result

```
linked list
    1.creation
    2.display
    3.insertbeg
    4.insertend
    5.insertpos
    6.dltbeg
    7.dltend
    8.dltpos
enter your choice :
1
enter the data 3

do you want to continue:(enter 1 for yes 0 for no)1
enter your choice :
4
enter data 5

do you want to continue:(enter 1 for yes 0 for no)1
enter your choice :
2
the elements are 3 5
do you want to continue:(enter 1 for yes 0 for no)1
enter your choice :
3
enter the data
7

do you want to continue:(enter 1 for yes 0 for no)1
enter your choice :
7
5is deleted
do you want to continue:(enter 1 for yes 0 for no)0

[Process completed - press Enter]
```

2.Polynomial Addition Using Array

```
#include<iostream>
#include<math.h>
using namespace std;
int main()
{
int a[10],b[10],c[10],n,x,y,i,s=0;
cout<<"enter highest degree of first polynomial : ";
cin>>x;
cout<<"enter highest degree of second polynomial : ";
cin>>y;
if(x>y)
{
n=x;
}
else if(x<y)
{
n=y;
}
else
{
n=x;
}
```

```

for(i=0;i<=n;i++)
{
a[i]=0;
}
for(i=0;i<=n;i++)
{
b[i]=0;
}
cout<<"first polynomial:\n";
for(i=0;i<=x;i++)
{
cout<<"enter coefficient of "<<i<<" degree : ";
cin>>a[i];
cout<<"\n";
}
cout<<"second polynomial:\n";
for(i=0;i<=y;i++)
{
cout<<"enter coefficient of "<<i<<" degree : ";
cin>>b[i];
cout<<"\n";
}
for(i=0;i<=n;i++)
{
c[i]=a[i]+b[i];
}

```

```

for(i=n;i>=0;i--)
{
cout<<c[i]<<"x^"<<i;
if(i==0)
{
break;
}
cout<<" ";
}
cout<<"\nenter value of x : ";
int z;
cin>>z;
for(i=n;i>=0;i--)
{
s=s+c[i]*pow(z,i);
}
cout<<"sum="<<s<<"\n";
}

```

OUTPUT

Compile Result

```
enter the degree of first polynomial :
3
enter the coefficients of first polynomial exp=0:
2
enter the coefficients of first polynomial exp=1:
3
enter the coefficients of first polynomial exp=2:
4
enter the coefficients of first polynomial exp=3:
5
enter the degree of second polynomial:
2
enter the coefficients of second polynomial exp=0:
3
enter the coefficients of second polynomial exp=1:
5
enter the coefficients of second polynomial exp=2:
6

result5x^3+10x^2+8x^1+5
[Process completed - press Enter]
```

3.Sparse Matrix Addition Using Array

```
#include<iostream>
using namespace std;
int main()
{
int sparse1[10][3],sparse2[10][3],sum[10][3];
int m,n,p,q,t1,t2,s,element;
int i,j;
cout<<"Enter the number of rows and columns : ";
cin>>m>>n;
t1=t2=0;
cout<<"Enter the first matrix("<<m<<"*"<<n<<"):";
for(i=1;i<=m;i++)
{
for(j=1;j<=n;j++)
{
cin>>element;
if(element!=0)
{
t1=t1+1;
sparse1[t1][1]=i;
sparse1[t1][2]=j;
sparse1[t1][3]=element;
}
}
}
sparse1[0][1]=m;
sparse1[0][2]=n;
sparse1[0][3]=t1;
cout<<"Enter the second matrix("<<m<<"*"<<n<<"):";
for(i=1;i<=m;i++)
{
for(j=1;j<=n;j++)
{
cin>>element;
if(element!=0)
```

```

{
t2=t2+1;
sparse2[t2][1]=i;
sparse2[t2][2]=j;
sparse2[t2][3]=element;
}
}
}
sparse2[0][1]=m;
sparse2[0][2]=n;
sparse2[0][3]=t2;
cout<<"The first sparse matrix is :\nRow Column Element\n";
cout<<"-----\n";
for(i=0;i<=t1;i++)
{
cout<<sparse1[i][1]<<" "<<sparse1[i][2]<<" "<<sparse1[i][3]<<"";
cout<<"\n";
}
cout<<"The second sparse matrix is :\nRow Column Element\n";
cout<<"-----\n";
for(i=0;i<=t2;i++)
{
cout<<sparse2[i][1]<<" "<<sparse2[i][2]<<" "<<sparse2[i][3]<<"";
cout<<"\n";
}
i=j=s=1;
while((i<=t1)&&(j<=t2))
{
if(sparse1[i][1]==sparse2[j][1])
{
if(sparse1[i][2]==sparse2[j][2])
{
sum[s][1]=sparse1[i][1];
sum[s][2]=sparse1[i][2];
sum[s][3]=sparse1[i][3]+sparse2[j][3];
i++;
j++;
if(sum[s][3]!=0)

```

```

s++;
}
else
{
if(sparse1[i][2]<sparse2[j][2])
{
sum[s][1]=sparse1[i][1];
sum[s][2]=sparse1[i][2];
sum[s][3]=sparse1[i][3];
i++;
s++;
}
else
{
sum[s][1]=sparse2[j][1];
sum[s][2]=sparse2[j][2];
sum[s][3]=sparse2[j][3];
j++;
s++;
}
}
}
else
{
if(sparse1[i][1]<sparse2[j][1])
{
sum[s][1]=sparse1[i][1];
sum[s][2]=sparse1[i][2];
sum[s][3]=sparse1[i][3];
i++;
s++;
}
else
{
sum[s][1]=sparse2[j][1];
sum[s][2]=sparse2[j][2];
sum[s][3]=sparse2[j][3];
j++;
s++;
}
}
}

```

```

}
}
}
if(i<=t1)
{
for(p=i;p<=t1;p++)
{
sum[s][1]=sparse1[p][1];
sum[s][2]=sparse1[p][2];
sum[s][3]=sparse1[p][3];
s++;
}
}
else if(j<=t2)
{
for(p=j;p<=t2;p++)
{
sum[s][1]=sparse2[p][1];
sum[s][2]=sparse2[p][2];
sum[s][3]=sparse2[p][3];
s++;
}
sum[0][1]=m;
sum[0][2]=n;
sum[0][3]=s-1;
cout<<"The sum is :\nRow Column Element\n";
cout<<"-----\n";
for(i=0;i<s;i++)
{
cout<<sum[i][1]<<" "<<sum[i][2]<<" "<<sum[i][3]<<"";
cout<<"\n";
}
}
return 0;
}

```

OUTPUT

Compile Result

```
Enter the number of rows and columns : 2
3
Enter the first matrix(2*3):1
2
3
0
4
0
Enter the second matrix(2*3):2
0
3
0
4
1
The first sparse matrix is :
Row Column Element
-----
2 3 4
1 1 1
1 2 2
1 3 3
2 2 4
The second sparse matrix is :
Row Column Element
-----
2 3 4
1 1 2
1 3 3
2 2 4
2 3 1
```

```
The sum is :  
Row Column Element
```

```
-----
```

```
2 3 5  
1 1 3  
1 2 2  
1 3 6  
2 2 8  
2 3 1
```

```
[Process completed - press Enter]
```

4.Stack Using Linked List

```
#include<iostream>
using namespace std;
class node
{
public:
int data;
node *next;
};
class stack
{
private:
node *temp,*fresh,*top;
public:
stack()
{
top=NULL;
}
void push()
{
fresh=new node();
cout<<"enter data to insert :\n";
cin>>fresh->data;
fresh->next=top;
top=fresh;
}
void display()
{
```

```

temp=top;
if(top==NULL)
{
cout<<"nothing to display , list is empty";
}
else
{
while(temp!=NULL)
{
cout<<temp->data<<" ";
temp=temp->next;
}
}
}

void pop()
{
node *temp;
temp=top;
if(top==NULL)
{
cout<<"underflow";
}
else
{
cout<<temp->data<<"is deleted";
temp=temp->next;
delete temp;
}
}

```

```

};
int main()
{
stack st;
int ch1,ch2;
do
{
cout<<"stack using linkedlist";
cout<<"enter your choice : \n1.push\n2.display\n3.pop";
cin>>ch1;
switch(ch1)
{
case 1:
st.push();
break;
case 2:
st.display();
break;
case 3:
st.pop();
break;
default:
cout<<"wrong entry";
}
cout<<"do you want to continue:(enter 1 for yes , 0 for no)";
cin>>ch2;
}while(ch2);
}

```

OUTPUT

Compile Result

```
stack using linkedlist
```

```
1.push
```

```
2.display
```

```
3.pop
```

```
Enter your choice 1
```

```
enter data to insert :
```

```
3
```

```
do you want to continue:(enter 1 for yes , 0 for no)1
```

```
Enter your choice 1
```

```
enter data to insert :
```

```
4
```

```
do you want to continue:(enter 1 for yes , 0 for no)1
```

```
Enter your choice 2
```

```
4
```

```
3
```

```
do you want to continue:(enter 1 for yes , 0 for no)1
```

```
Enter your choice3
```

```
4is deleted
```

```
do you want to continue:(enter 1 for yes , 0 for no)
```

5.Queue Operation Using Array

```
#include<iostream>
using namespace std;
class queue
{
private:
int *arr,n;
int front,rear;
public:
queue()
{
arr=new int[n];
front=-1;
rear=-1;
}
void enqueue()
{
int item;
cout<<"enter the item :\n";
cin>>item;
if(rear==n-1)
{
cout<<"queue overflow , item can't insert";
return;
}
else if(front==n-1)
{
```

```

front++;
rear++;
arr[rear]=item;
}
else
{
rear++;
arr[rear]=item;
}
}
void display()
{
int i;
if(front==-1||front>rear)
{
cout<<"q is empty";
}
else
{
for(i=front;i<=rear;i++)
{
cout<<arr[i]<<" ";
}
}
}
void deque()
{
if(front==-1||front>rear)
{

```

```

cout<<"queue underflow ,item can't insert";
return;
}
else if(front==rear)
{
cout<<arr[front]<<"is deleted";
front=rear=-1;
}
else
{
cout<<arr[front]<<"is deleted";
front++;
}
}
};
int main()
{
    queue q;
    int ch1,ch2;
    do
    {
        cout<<"queue using array\n";
        cout<<"\nenter your choice : \n1.insertion\n2.display\n3.deletion\n";
        cin>>ch1;
        switch(ch1)
        {
            case 1:
                q.enqueue();
                break;
            case 2:

```

```
q.display();
break;
case 3:
q.dequeue();
break;
default:
cout<<"wrong entry\n";
}
cout<<"\ndo you want to continue:(enter 1 for yes , 0 for no)\n";
cin>>ch2;
}while(ch2);
}
```

OUTPUT

Compile Result

```
queue using array
```

```
1.insertion  
2.display  
3.deletion
```

```
enter your choice : 1  
enter the item :  
3
```

```
do you want to continue:(enter 1 for yes , 0 for no)  
1
```

```
enter your choice : 1  
enter the item :  
4
```

```
do you want to continue:(enter 1 for yes , 0 for no)  
1
```

```
enter your choice : 2
```

```
the elements are  
3 4
```

```
do you want to continue:(enter 1 for yes , 0 for no)  
1
```

```
enter your choice : 3  
3is deleted  
do you want to continue:(enter 1 for yes , 0 for no)  
0
```

```
[Process completed - press Enter]
```

6.Evaluation of Arithmetic Expression

```
#include<iostream>
#include<cmath>
#include<stack>
using namespace std;
float scanNum(char ch)
{
int value;
value = ch;
return float(value-'0');
}
int isOperator(char ch)
{
if(ch == '+' || ch == '-' || ch == '*' || ch == '/' || ch == '^')
return 1;
return -1;
}
int isOperand(char ch)
{
if(ch >= '0' && ch <= '9')
return 1;
return -1;
29}
float operation(int a, int b, char op)
{
if(op == '+')
return b+a;
```

```

else if(op == '-')
return b-a;
else if(op == '*')
return b*a;
else if(op == '/')
return b/a;
else if(op == '^')
return pow(b,a);
else
return(-1);
}
float posteval(string postfix)
{
int a, b;
stack<float> stk;
string::iterator it;
for(it=postfix.begin(); it!=postfix.end(); it++)
{
if(isOperator(*it) != -1)
{
a = stk.top();
stk.pop();
b = stk.top();
stk.pop();
stk.push(operation(a, b, *it));
}
else if(isOperand(*it) > 0)
{
stk.push(scanNum(*it));
}
}
}

```

```
}  
return stk.top();  
}  
int main()  
{  
string post;  
cout<<"enter post fix expression\n";  
cin>>post;  
cout<<"the result is:"<<posteval(post);  
}
```

OUTPUT

Compile Result

```
Enter post fix expression:  
53*2+6  
The result is:6  
[Process completed - press Enter]
```

7.BST Operation

```
#include<iostream>
#include<stdlib.h>
using namespace std;
struct treeNode
{
int data;
treeNode *left;
treeNode *right;
};
treeNode *Insert(treeNode *node,int data)
{
if(node==NULL)
{
treeNode *temp;
temp=new treeNode;
temp -> data = data;
temp -> left = temp -> right = NULL;
return temp;
}
if(data >(node->data))
{
node->right = Insert(node->right,data);
}
else if(data < (node->data))
{
node->left = Insert(node->left,data);
}
```

```

}
return node;
}
treeNode * Delet(treeNode *node, int data)
{
treeNode *temp;
if(node==NULL)
{
cout<<"Element Not Found";
}
else if(data < node->data)
{
node->left = Delet(node->left, data);
}
else if(data > node->data)
{
node->right = Delet(node->right, data);
}
else
{
if(node->right && node->left)
{
node -> data = temp->data;
node -> right = Delet(node->right,temp->data);
}
else
{
temp = node;
if(node->left == NULL)

```

```

node = node->right;
else if(node->right == NULL)
node = node->left;
free(temp);
}
}
return node;
}
treeNode * Find(treeNode *node, int data)
{
if(node==NULL)
{
return NULL;
}
if(data > node->data)
{
return Find(node->right,data);
}
else if(data < node->data)
{
return Find(node->left,data);
}
else
{
return node;
}
}
void Inorder(treeNode *node)
{
if(node==NULL)

```

```

{
return;
}
Inorder(node->left);
cout<<node->data<<" ";
Inorder(node->right);
}
void Preorder(treeNode *node)
{
if(node==NULL)
{
return;
}
cout<<node->data<<" ";
Preorder(node->left);
Preorder(node->right);
}
void Postorder(treeNode *node)
{
if(node==NULL)
{
return;
}
Postorder(node->left);
Postorder(node->right);
cout<<node->data<<" ";
}
int main()
{

```

```

treeNode *root = NULL,*temp;
int ch,ch2;
do
{
cout<<"\n1.Insert\n2.Delete\n3.Inorder\n4.Preorder\n5.Postorder\n6.Search\n7.Exit\n";
cout<<"Enter ur choice:";
cin>>ch;
switch(ch)
{
case 1:
cout<<"\nEnter element to be insert:";
cin>>ch;
root = Insert(root, ch);
cout<<"\nElements in BST are:";
Inorder(root);
break;
case 2:
cout<<"\nEnter element to be deleted:";
cin>>ch;
root = Delet(root,ch);
cout<<"\nAfter deletion elements in BST are:";
Inorder(root);
break;
case 3:
cout<<"\nInorder Travesals is:";
Inorder(root);
break;
case 4:
cout<<"\nPreorder Traversals is:";

```

```

Preorder(root);
break;
case 5:
cout<<"\nPostorder Traversals is:";
Postorder(root);
break;
case 6:
cout<<"\nEnter element to be searched:";
cin>>ch;
temp = Find(root,ch);
if(temp==NULL)
{
cout<<"Element is not found\n";
}
else
{
cout<<"Element "<<temp->data<<" is Found\n";
}
break;
case 7:
exit(0);
break;
default:
cout<<"\nEnter correct choice:";
break;
}
}while(ch2);
return 0;
}

```

OUTPUT

Compile Result

```
1.Insert
2.Delete
3.Inorder
4.Preorder
5.Postorder
6.Search
7.Exit
```

```
Enter ur choice:1
```

```
Enter element to be insert:2
```

```
Elements in BST are:2
Enter ur choice:1
```

```
Enter element to be insert:4
```

```
Elements in BST are:2 4
Enter ur choice:1
```

```
Enter element to be insert:7
```

```
Elements in BST are:2 4 7
Enter ur choice:1
```

```
Enter element to be insert:9
```

```
Elements in BST are:2 4 7 9
Enter ur choice:3
```

```
Inorder Travesals is:2 4 7 9
Enter ur choice:4
```

```
Preorder Traversals is:2 4 7 9
```

Enter ur choice:6

Enter element to be searched:4
Element 4 is Found

Enter ur choice:7

[Process completed - press Enter]

8.Heap Sort

```
#include<iostream>
using namespace std;
void heap(int a[],int n,int i)
{
    int d=i;
    int l=2*i+1;
    int r=2*i+2;
    if(l<n && a[l]>a[d])
        d=l;
    if(r<n && a[r]>a[d])
    {
        d=r;
    }
    if(d!=i)
    {
        swap(a[i],a[d]);
        heap(a,n,d);
    }
}
void heapsrt(int a[],int n)
{
    for(int i=n/2-1;i>=0;i--)
    {
        heap(a,n,i);
    }
    for(int i=n-1;i>0;i--)
    {

```

```

swap(a[0],a[i]);
heap(a,i,0);
}
}
int main()
{
int n,a[20],i;
cout<<"enter the number of elements:\n";
cin>>n;
cout<<"enter the elements :\n";
for(i=0;i<n;i++)
{
cin>>a[i];
}
cout<<"entered elements are :\n";
for(i=0;i<n;i++)
{
cout<<a[i]<<" ";
}
heapsrt(a,n);
cout<<" sorted array is :\n";
for(i=0;i<n;i++)
{
cout<<a[i]<<" ";
}
}

```

OUTPUT

Compile Result

```
enter the number of elements:  
5  
enter the elements :  
2  
9  
4  
3  
6  
entered elements are :  
2 9 4 3 6  
  sorted array is :  
2 3 4 6 9  
[Process completed - press Enter]
```

9.Hashing

```
#include<iostream>
#include<cstdlib>
#include<string>
#include<cstdio>
using namespace std;
const int T_S = 200;
class HashTableEntry {
public:
int k;
int v;
HashTableEntry(int k, int v) {
this->k= k;
this->v = v;
}
};
class HashMapTable
{
private:
    HashTableEntry **t;
public:
    HashMapTable() {
t = new HashTableEntry * [T_S];
for (int i = 0; i< T_S; i++)
    {
        t[i] = NULL;
    }
}
```

```

}
int HashFunc(int k)
{
return k % T_S;
}
void Insert(int k, int v)
{
int h = HashFunc(k);
while (t[h] != NULL && t[h]->k != k)
{
h = HashFunc(h + 1);
}
if (t[h] != NULL)
delete t[h];
t[h] = new HashTableEntry(k, v);
}
int SearchKey(int k)
{
int h = HashFunc(k);
while (t[h] != NULL && t[h]->k != k)
{
h = HashFunc(h + 1);
}
if (t[h] == NULL)
return -1;
else
return t[h]->v;
}
void Remove(int k)
{

```

```

int h = HashFunc(k);
while (t[h] != NULL)
{
    if (t[h]->k == k)
        break;
    h = HashFunc(h + 1);
}
if (t[h] == NULL)
{
    cout<<"No Element found at key "<<k<<endl;
    return;
}
else
{
    delete t[h];
}
cout<<"Element Deleted"<<endl;
}
~HashMapTable()
{
    for (int i = 0; i < T_S; i++)
    {
        if (t[i] != NULL)
            delete t[i];
        delete[] t;
    }
}
};

int main()

```

```

{
HashMapTable hash;
int k, v;
int c;
while (1)
{
cout<<"1.Insert element into the table"<<endl;
cout<<"2.Search element from the key"<<endl;
cout<<"3.Delete element at a key"<<endl;
cout<<"4.Exit"<<endl;
cout<<"Enter your choice: ";
cin>>c;
switch(c)
{
case 1:
    cout<<"Enter element to be inserted: ";
    cin>>v;
    cout<<"Enter key at which element to be inserted: ";
    cin>>k;
    hash.Insert(k, v);
    break;
case 2:
    cout<<"Enter key of the element to be searched: ";
    cin>>k;
    if (hash.SearchKey(k) == -1)
    {
        cout<<"No element found at key "<<k<<endl;
        continue;
    }
    else

```

```

        {
            cout<<"Element at key "<<k<<" : ";
            cout<<hash.SearchKey(k)<<endl;
        }
        break;
case 3:
    cout<<"Enter key of the element to be deleted: ";
    cin>>k;
    hash.Remove(k);
    break;
case 4:
    exit(1);
default:
    cout<<"\nEnter correct option\n";
    }
}
return 0;
}

```

OUTPUT

Compile Result

```
1.Insert element into the table
2.Search element from the key
3.Delete element at a key
4.Exit
Enter your choice: 1
Enter element to be inserted: 3
Enter key at which element to be inserted: 2
1.Insert element into the table
2.Search element from the key
3.Delete element at a key
4.Exit
Enter your choice: 1
Enter element to be inserted: 2
Enter key at which element to be inserted: 4
1.Insert element into the table
2.Search element from the key
3.Delete element at a key
4.Exit
Enter your choice: 2
Enter key of the element to be searched: 4
Element at key 4 : 2
1.Insert element into the table
2.Search element from the key
3.Delete element at a key
4.Exit
Enter your choice: 3
Enter key of the element to be deleted: 4
Element Deleted
1.Insert element into the table
2.Search element from the key
3.Delete element at a key
4.Exit
Enter your choice: 4

[Process completed (code 1) - press Enter]
```

10.Binary Search

```
#include<iostream>
using namespace std;
int main()
{
int i,beg,end,mid,n,item,a[30];
cout<<"enter the number of element :\n";
cin>>n;
cout<<"enter the elements :\n";
for(i=0;i<n;i++)
{
cin>>a[i];
}
cout<<"enter the item to be searched :\n";
cin>>item;
cout<<"the elements are \n";
for(i=0;i<n;i++)
{
cout<<a[i]<<" ";
}
beg=0;
end=n-1;
mid=(beg+end)/2;
while(a[mid]!=item && beg<end)
{
if(a[mid]>item)
{
end=mid-1;
```

```
}  
else  
{  
    beg=mid+1;  
    mid=(beg+end)/2;  
}  
}  
if(a[mid]==item)  
{  
    cout<<"item found at the position:\n"<<mid+1;  
}  
else  
{  
    cout<<" item not found:\n";  
}  
}
```

OUTPUT

Compile Result

```
enter the number of element :  
5  
enter the element :  
2  
4  
5  
7  
8  
enter the item to be searched :  
7  
  
the elements are  
2 4 5 7 8  
item found at the position:  
4  
[Process completed - press Enter]
```

11.Quick Sort

```
#include<iostream>
using namespace std;
int partition(int a[ ],int lb,int ub)
{
int pivot=a[lb];
int count=0;
for(int i=lb+1;i<=ub;i++)
{
if(a[i]<=pivot)
{
count++;
}
}
int loc=lb+count;
swap(a[loc],a[lb]);
int i=lb,j=ub;
while(i<loc && j>loc)
{
while( a[i]<=pivot)
{
i++;
}
while(a[j]>pivot)
{
j--;
```

```

}
if(i<loc && j>loc)
{
swap(a[i++],a[j--]);
}
}
return loc;
}
void quicksort(int a[ ],int lb,int ub)
{
if(lb>=ub)
return;
int loc=partition(a,lb,ub);
quicksort(a,lb,loc-1);
quicksort(a,loc+1,ub);
}
int main()
{
int n,a[20];
cout<<"enter the limit:\n";
cin>>n;
cout<<"enter the elements:\n";
for(int i=0;i<n;i++)
{
cin>>a[i];
}
quicksort(a,0,n-1);
for(int i=0;i<n;i++)
{

```

```
cout<<a[i]<<" ";  
}  
}
```

OUTPUT

Compile Result

```
enter the limit:  
5  
enter the elements:  
1 8 3 6 2  
1 2 3 6 8  
[Process completed - press Enter]
```

12.N-queen Problem Using Backtracking Algorithm

```
#include<iostream>
using namespace std;
#define N 4
void printBoard(int board[N][N])
{
    for (int i = 0; i < N; i++)
    {
        for (int j = 0; j < N; j++)
            cout << board[i][j] << " ";
        cout << endl;
    }
}
bool isValid(int board[N][N], int row, int col)
{
    for (int i = 0; i < col; i++)
        if (board[row][i])
            return false;
    for (int i=row, j=col; i>=0 && j>=0; i--, j--)
        if (board[i][j])
            return false;
    for (int i=row, j=col; j>=0 && i<N; i++, j--)
        if (board[i][j])
            return false;
    return true;
}
```

```

bool solveNQueen(int board[N][N], int col)
{
    if (col >= N)
        return true;
    for (int i = 0; i < N; i++)
    {
        if (isValid(board, i, col) )
        {
            board[i][col] = 1;
            if ( solveNQueen(board, col + 1))
                return true;
            board[i][col] = 0;
        }
    }
    return false;
}

bool checkSolution()
{
    int board[N][N];
    for(int i = 0; i<N; i++)
        for(int j = 0; j<N; j++)
            board[i][j] = 0;
    if ( solveNQueen(board, 0) == false )
    {
        cout << "Solution does not exist";
        return false;
    }
    printBoard(board);
    return true;
}

```

```
int main()
{
    checkSolution();
}
```

OUTPUT

Compile Result

```
0 0 1 0
1 0 0 0
0 0 0 1
0 1 0 0
```

```
[Process completed - press Enter]
```

